Department	Hosts Needed	Nearest Subnet Size	Subnet Mask	Subnet
A	20	32 (5 bits)	/27	192.104.1.0/27
B	10	16 (4 bits)	/28	192.104.1.32/28
C	3	8 (3 bits)	/29	192.104.1.48/29

WAN Link	Subnet
R1 ↔ R2	192.104.1.56/30
R1 ↔ R3	192.104.1.60/30
R2 ↔ R3	192.104.1.64/30

Router1

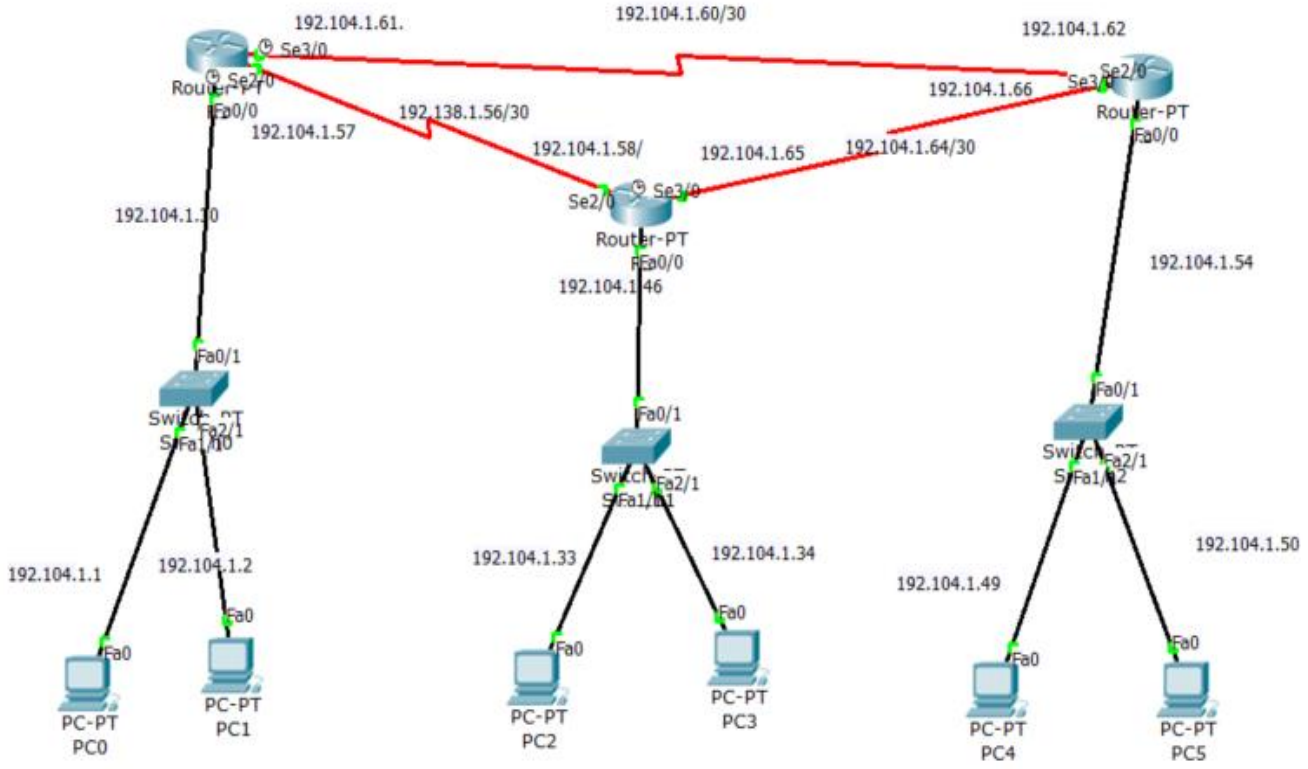
```
Router>enable
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.104.1.0/24 is variably subnetted, 5 subnets, 4 masks
C       192.104.1.0/27 is directly connected, FastEthernet0/0
S       192.104.1.32/28 [1/0] via 192.104.1.58
S       192.104.1.48/29 [1/0] via 192.104.1.62
C       192.104.1.56/30 is directly connected, Serial2/0
C       192.104.1.60/30 is directly connected, Serial3/0
Router#
```

Router2

```
Router>enable
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.104.1.0/24 is variably subnetted, 5 subnets, 4 masks
S       192.104.1.0/27 [1/0] via 192.104.1.57
C       192.104.1.32/28 is directly connected, FastEthernet0/0
S       192.104.1.48/29 [1/0] via 192.104.1.66
C       192.104.1.56/30 is directly connected, Serial2/0
C       192.104.1.64/30 is directly connected, Serial3/0
Router#
```

Router3

```
Router>enable
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.104.1.0/24 is variably subnetted, 5 subnets, 4 masks
S       192.104.1.0/27 [1/0] via 192.104.1.61
S       192.104.1.32/28 [1/0] via 192.104.1.65
C       192.104.1.48/29 is directly connected, FastEthernet0/0
C       192.104.1.60/30 is directly connected, Serial2/0
C       192.104.1.64/30 is directly connected, Serial3/0
Router#
```



Ping commands pc0

PC0

Physical | Config | Desktop | Custom Interface

Command Prompt

```
Packet Tracer PC Command Line 1.0
PC>ping 192.104.1.33

Pinging 192.104.1.33 with 32 bytes of data:

Reply from 192.104.1.33: bytes=32 time=1ms TTL=126
Reply from 192.104.1.33: bytes=32 time=2ms TTL=126
Reply from 192.104.1.33: bytes=32 time=8ms TTL=126
Reply from 192.104.1.33: bytes=32 time=1ms TTL=126

Ping statistics for 192.104.1.33:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 8ms, Average = 3ms

PC>ping 192.104.1.49

Pinging 192.104.1.49 with 32 bytes of data:

Reply from 192.104.1.49: bytes=32 time=2ms TTL=126
Reply from 192.104.1.49: bytes=32 time=1ms TTL=126
Reply from 192.104.1.49: bytes=32 time=6ms TTL=126
Reply from 192.104.1.49: bytes=32 time=11ms TTL=126

Ping statistics for 192.104.1.49:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 11ms, Average = 5ms

PC>ping 192.104.1.1

Pinging 192.104.1.1 with 32 bytes of data:

Reply from 192.104.1.1: bytes=32 time=9ms TTL=128
Reply from 192.104.1.1: bytes=32 time=0ms TTL=128
Reply from 192.104.1.1: bytes=32 time=8ms TTL=128
Reply from 192.104.1.1: bytes=32 time=9ms TTL=128

Ping statistics for 192.104.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 9ms, Average = 6ms

PC>
```

PC 3

PC3

Physical | Config | Desktop | Custom Interface

Command Prompt

```
Packet Tracer PC Command Line 1.0
PC>ping 192.104.1.1

Pinging 192.104.1.1 with 32 bytes of data:

Reply from 192.104.1.1: bytes=32 time=1ms TTL=126
Reply from 192.104.1.1: bytes=32 time=4ms TTL=126
Reply from 192.104.1.1: bytes=32 time=2ms TTL=126
Reply from 192.104.1.1: bytes=32 time=2ms TTL=126

Ping statistics for 192.104.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 4ms, Average = 2ms

PC>ping 192.104.1.50

Pinging 192.104.1.50 with 32 bytes of data:

Reply from 192.104.1.50: bytes=32 time=2ms TTL=126
Reply from 192.104.1.50: bytes=32 time=8ms TTL=126
Reply from 192.104.1.50: bytes=32 time=1ms TTL=126
Reply from 192.104.1.50: bytes=32 time=1ms TTL=126

Ping statistics for 192.104.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 8ms, Average = 3ms

PC>ping 192.104.1.33

Pinging 192.104.1.33 with 32 bytes of data:

Reply from 192.104.1.33: bytes=32 time=1ms TTL=128
Reply from 192.104.1.33: bytes=32 time=1ms TTL=128
Reply from 192.104.1.33: bytes=32 time=0ms TTL=128
Reply from 192.104.1.33: bytes=32 time=1ms TTL=128

Ping statistics for 192.104.1.33:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

PC>
```

PC4

PC4

Physical | Config | Desktop | Custom Interface

Command Prompt

```
Packet Tracer PC Command Line 1.0
PC>ping 192.104.1.34

Pinging 192.104.1.34 with 32 bytes of data:

Reply from 192.104.1.34: bytes=32 time=7ms TTL=126
Reply from 192.104.1.34: bytes=32 time=1ms TTL=126
Reply from 192.104.1.34: bytes=32 time=4ms TTL=126
Reply from 192.104.1.34: bytes=32 time=1ms TTL=126

Ping statistics for 192.104.1.34:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 7ms, Average = 3ms

PC>ping 192.104.1.50

Pinging 192.104.1.50 with 32 bytes of data:

Reply from 192.104.1.50: bytes=32 time=0ms TTL=128
Reply from 192.104.1.50: bytes=32 time=0ms TTL=128
Reply from 192.104.1.50: bytes=32 time=1ms TTL=128
Reply from 192.104.1.50: bytes=32 time=0ms TTL=128

Ping statistics for 192.104.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

PC>ping 192.104.1.1

Pinging 192.104.1.1 with 32 bytes of data:

Reply from 192.104.1.1: bytes=32 time=1ms TTL=126
Reply from 192.104.1.1: bytes=32 time=7ms TTL=126
Reply from 192.104.1.1: bytes=32 time=1ms TTL=126
Reply from 192.104.1.1: bytes=32 time=2ms TTL=126

Ping statistics for 192.104.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 7ms, Average = 2ms

PC>
```

Router 1

R1

Physical Config CLI

```
PT2005 processor: part number 0, mask 01
Bridging software.
X.25 software, Version 3.0.0.
4 FastEthernet/IEEE 802.3 interface(s)
2 Low-speed serial(sync/async) network interface(s)
32K bytes of non-volatile configuration memory.
63488K bytes of ATA CompactFlash (Read/Write)


         --- System Configuration Dialog ---

Continue with configuration dialog? [yes/no]: NO


Press RETURN to get started!



Router>ENABLE
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fa0/0
Router(config-if)#ip address 192.104.1.57 255.255.255.224
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state t
o up

Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#ip address 192.104.1.57 255.255.255.252
% 192.104.1.56 overlaps with FastEthernet0/0
Router(config-if)#exit
Router(config)#no ip address
                     ^
% Invalid input detected at '^' marker.
```

```
Router(config)#interface FastEthernet0/0
Router(config-if)#no ip address
Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#ip address 192.104.1.57 255.255.255.252
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial2/0, changed state to down
Router(config-if)#exit
Router(config)#interface Serial3/0
Router(config-if)#ip address 192.104.1.61 255.255.255.252
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.104.1.30 255.255.255.224
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#ip route 192.104.1.32 255.255.255.240 192.104.1.58
Router(config)#ip route 192.104.1.48 255.255.255.248 192.104.1.60
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write
Building configuration...
[OK]
Router#









Router con0 is now available





Press RETURN to get started.


Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#no ip route 192.104.1.48 255.255.255.248 192.104.1.60
Router(config)#ip route 192.104.1.48 255.255.255.248 192.104.1.62
Router(config)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

%LINK-5-CHANGED: Interface Serial3/0, changed state to up
```

Router2

R2

Physical Config CLI

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.104.1.46 255.255.255.240
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state t
o up

Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#ip address 192.104.1.58 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config-if)#exit
Router(config)#interface Serial3/0
Router(config-if)#ip address 192.104.1.65 255.255.255.252
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#exit
Router(config)#ip route 192.104.1.0 255.255.255.224 192.104.1.57
Router(config)#ip route 192.104.1.48 255.255.255.248 192.104.1.66
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write
Building configuration...
[OK]
Router#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up









Router con0 is now available




Press RETURN to get started.
```

Router3

R3

Physical Config CLI

IOS Command Line Inter

```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.104.1.54 255.255.255.248
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state t
o up

Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#ip address 192.104.1.62 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config-if)#exit
Router(config)#interface se3/0
Router(config-if)#ip address 192.104.1.66 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up

Router(config-if)#
Router(config-if)#exit
Router(config)#ip route 192.104.1.0 255.255.255.224 192.104.1.61
Router(config)#ip route 192.104.1.32 255.255.255.240 192.104.1.65
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write
Building configuration...
[OK]
Router#
```

```
Router con0 is now available
```

```
Press RETURN to get started.
```